

Industrial Internship Report on, "Multi-Domain Predictive Analytics Using Machine Learning Regression Models"

Crop Production, Mining Quality, Traffic Volume & Turbofan Engine RUL Forecasting

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Executive Summary

This report presents the work completed during the Data Science and Machine Learning Industrial Internship provided by upSkill Campus and The IoT Academy, in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

The internship was focused on applied Machine Learning using Python, and was completed over four weeks, each week centered around one independent, real-world regression problem.

My project work involved building four independent Machine Learning models — Crop Production Prediction, Mining Ore Quality (Silica Concentrate) Prediction, Traffic Volume Forecasting, and Turbofan Engine Remaining Useful Life (RUL) Prediction — each developed in Google Colab using Random Forest Regression, and evaluated using R^2 Score, MAE, and RMSE.

This internship gave me strong, hands-on exposure to the complete Machine Learning workflow — from raw data ingestion and preprocessing to model training, evaluation, and visualization — and was an overall great learning experience.

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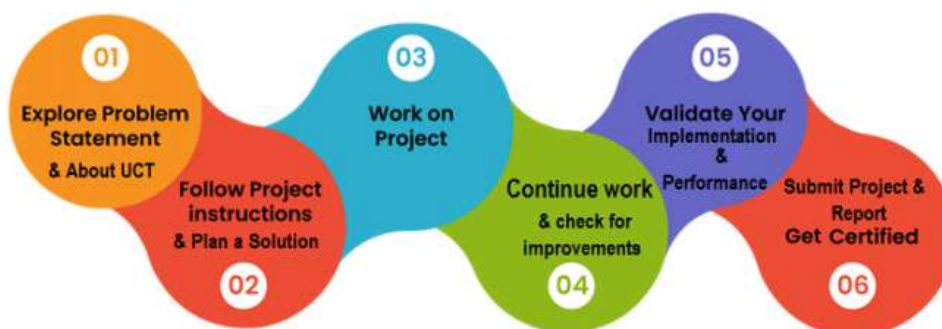
1. Preface

This report summarizes the work completed during the four-week Data Science and Machine Learning Industrial Internship organized by upSkill Campus (USC) and The IoT Academy, in collaboration with UniConverge Technologies Pvt. Ltd. (UCT).

Internships play a vital role in a student's career development. They bridge the gap between classroom learning and industry practice, allowing students to apply theoretical knowledge to real, hands-on project work while building confidence, discipline, and problem-solving ability that are difficult to gain from academics alone.

During this internship, I worked on four independent Machine Learning regression projects — Crop Production Prediction, Mining Quality (Silica Concentrate) Prediction, Traffic Forecasting, and Turbofan Engine Remaining Useful Life (RUL) Prediction. Each project was completed over one week, with a structured weekly progress report documenting the objectives, achievements, challenges faced, and lessons learned.

The program was planned as a weekly, project-based learning track: each week introduced a new real-world dataset and problem statement, which I solved independently using Python and Scikit-learn in Google Colab, evaluated thoroughly using regression metrics, and documented. This iterative structure helped me steadily build proficiency, moving from basic data cleaning and a single tabular dataset in Week 1 to a large industrial sensor dataset (737,453 rows) in Week 2, and a multivariate time-series sensor dataset (NASA C-MAPSS) in Week 4.



Over the course of this internship, I gained practical experience in data preprocessing, feature engineering, exploratory data analysis (EDA), Random Forest Regression, and model evaluation. Beyond the technical skills, I also improved my logical thinking, debugging discipline, and ability to independently research and resolve issues — all of which are directly relevant to a career in Data Science and Machine Learning.

I would like to thank upSkill Campus, The IoT Academy, and UniConverge Technologies Pvt. Ltd. for structuring and providing this internship opportunity, and my faculty at Mangayarkarasi College of Engineering for their continued support and encouragement throughout this program.

2. Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013, working in the Digital Transformation domain and providing Industrial solutions with a prime focus on sustainability and RoI.

For developing its products and solutions, UCT leverages various cutting-edge technologies such as Internet of Things (IoT), Cyber Security, Cloud Computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, and Front-end technologies.



- UCT Insight — an IoT platform for quick deployment of IoT applications with device connectivity via MQTT, CoAP, HTTP, Modbus TCP, and OPC UA, supporting cloud and on-premises deployments.
- Factory Watch — a Smart Factory platform providing scalable production and asset monitoring, OEE and predictive maintenance solutions.

- LoRaWAN-based Solutions — for Agritech, Smart Cities, Industrial Monitoring, and Smart Metering.
- Predictive Maintenance — Industrial machine health monitoring using Embedded Systems, Industrial IoT, and Machine Learning to estimate Remaining Useful Life of machines.

2.2 About upSkill Campus (USC)

upSkill Campus, along with The IoT Academy and in association with UniConverge Technologies, facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers personalized executive coaching in a more affordable, scalable, and measurable way, aiming to upskill 1 million learners over the next 5 years through career growth/upskilling courses, professional networking, a collaboration platform, and a job/internship platform.



2.3 The IoT Academy

The IoT Academy is the EdTech division of UCT that runs long executive certification programs in collaboration with EICT Academy, IITK, IITR, and IITG across multiple domains.

2.4 Objective

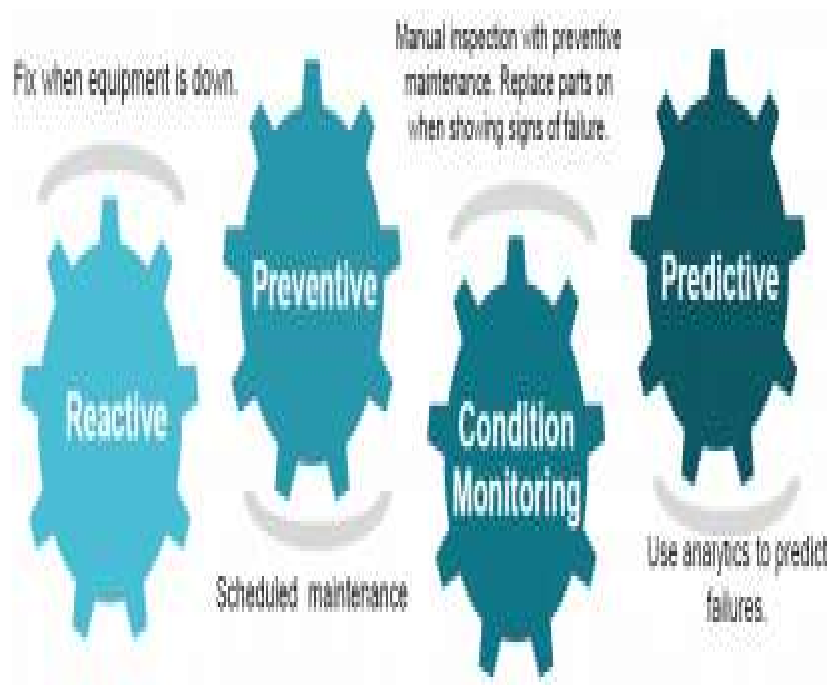
The objective of this internship program was to:

- Get practical, hands-on experience of applying Machine Learning in the software/data industry.
- Solve real-world, problem-statement-driven Data Science tasks across different domains — agriculture, mining, transportation, and predictive maintenance.
- Improve job prospects through demonstrable, tested project work.
- Gain an improved understanding of the end-to-end Machine Learning workflow using Python.
- Support personal growth in communication, documentation, and problem-solving ability.



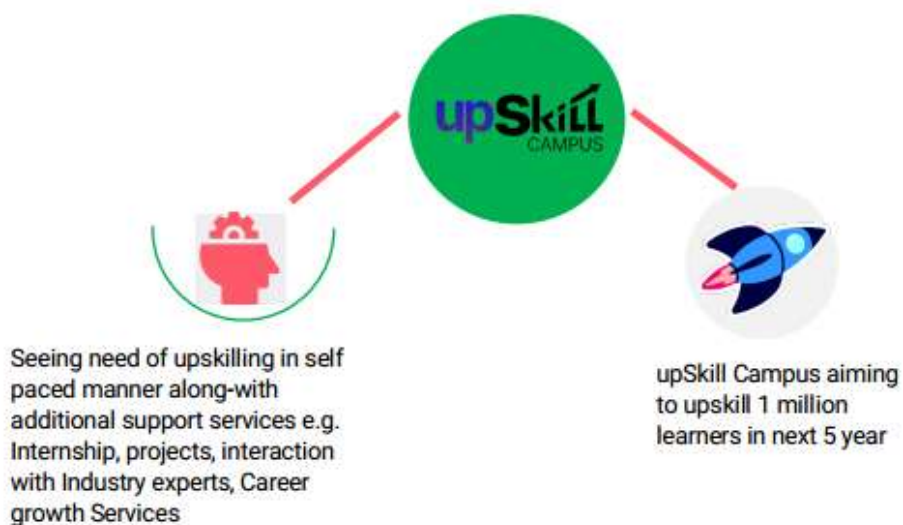
2.5 Reference

- [1] Scikit-learn Documentation, <https://scikit-learn.org/stable/>
- [2] Pandas Documentaton, <https://pandas.pydata.org/docs/>
- [3] NASA C-MAPSS Turbofan Engine Degradation Simulation Dataset, NASA Prognostics Data Repository
- [4] GeeksforGeeks — Machine Learning Tutorials, <https://www.geeksforgeeks.org/machine-learning/>



2.6 Glossary

Term	Meaning
ML	Machine Learning
EDA	Exploratory Data Analysis
RUL	Remaining Useful Life



Term	Meaning
R^2	Coefficient of Determination (R-squared Score)
MAE	Mean Absolute Error
RMSE	Root Mean Squared Error
CSV	Comma-Separated Values

3. Problem Statement

In real-world domains such as agriculture, mining, urban transportation, and industrial equipment maintenance, several important quantities need to be forecast or estimated in advance, but this is often done manually, inconsistently, or not at all:

- Agricultural production planning is often based on historical trends estimated manually, without a data-driven model to predict crop production from year, crop type, and seasonal features.
- Mining plants generate very large volumes of sensor data (flotation, flow, pH, density) but the resulting silica concentrate quality is typically measured after the fact through lab testing rather than predicted in real time from process sensor readings.
- Urban traffic volume at road junctions is difficult to anticipate in advance, making it hard for city planners to proactively manage congestion without a predictive model based on time and date features.
- Industrial machines such as turbopfan engines degrade gradually, and without a predictive model, their Remaining Useful Life (RUL) is not known in advance, increasing the risk of unplanned failure or unnecessarily early maintenance.

The assigned problem statement for this internship was to design and build independent, dataset-driven Machine Learning regression models that solve each of these four real-world prediction problems using Python, Pandas, and Scikit-learn, following a complete workflow of data preprocessing, feature engineering, model training, and performance evaluation.

4. Existing and Proposed Solution

Existing Solutions and Their Limitations

- Crop production estimates are often based on manual historical trend analysis by agricultural departments, without a systematic, data-driven predictive model.
- Mining plants rely on periodic manual/lab-based quality testing of silica concentrate rather than continuous, sensor-driven prediction, causing delays in process correction.
- Traffic management systems in many cities react to congestion as it happens rather than forecasting vehicle volume in advance from time-based patterns.
- Predictive maintenance for industrial engines is frequently either purely reactive (fix after failure) or based on fixed preventive schedules, rather than a data-driven Remaining Useful Life estimate.

Proposed Solution

Four independent Machine Learning regression models were designed and developed in Google Colab, each addressing one problem statement directly using Random Forest Regression:

- Crop Production Prediction — a Random Forest Regression model trained on a wide-format agricultural time-series dataset (reshaped using pandas melt), predicting crop production from Particulars, Frequency, Unit, and Year.

- Mining Quality Prediction — a Random Forest Regression model trained on the Mining Process Flotation Plant sensor dataset (737,453 records), predicting % Silica Concentrate from flow, pH, density, and air-flow sensor readings.
- Traffic Forecasting — a Random Forest Regression model trained on junction-wise traffic data, using engineered time features (Hour, Day, Month, Year, Weekday, IsWeekend) to predict vehicle volume.
- Turbofan RUL Prediction — a Random Forest Regression model trained on the NASA C-MAPSS turbofan engine sensor dataset, predicting Remaining Useful Life (RUL) from operating settings and sensor readings.

4.1 Code Submission (GitHub link)

<https://github.com/asumayafrin/upskillcampus>

4.2 Report Submission (GitHub link)

<https://github.com/asumayafrin/upskillcampus/tree/main/DataScience-ML-Projects>

5. Proposed Design / Model

All four Machine Learning projects follow the same overall design workflow: raw data is ingested (CSV/TXT upload in Google Colab), cleaned and preprocessed, relevant features are engineered or selected, a Random Forest Regression model is trained on an 80/20 train-test split, and the model's predictions are evaluated and visualized against actual values.

5.1 High Level Diagram

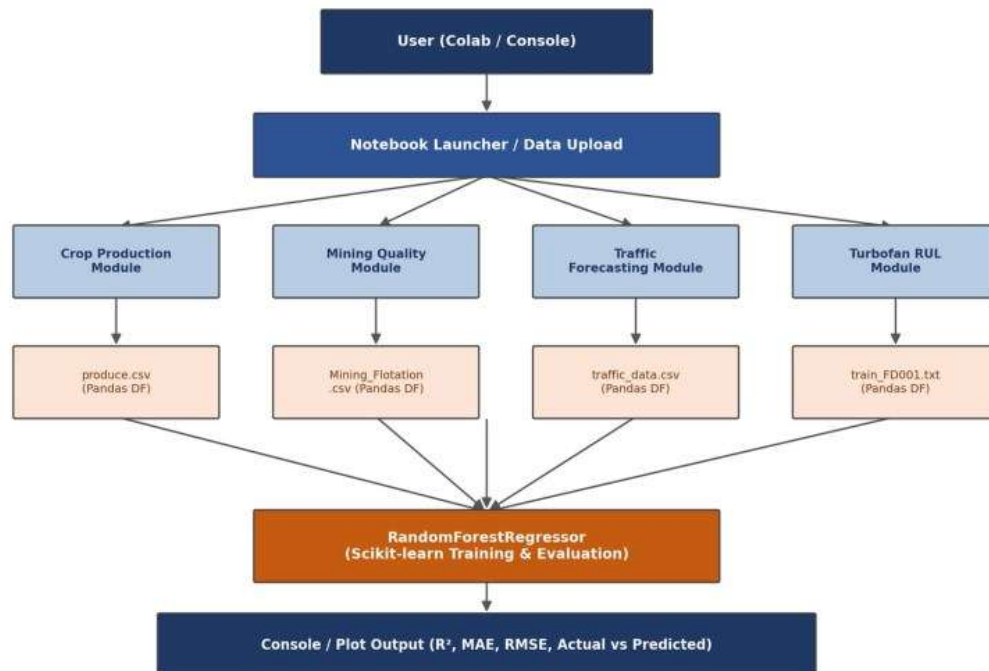


Figure 1: High-Level Architecture of the ML Prediction Suite

The Notebook Launcher acts as the entry point that directs execution to one of the four independent modules. Each module loads its own dataset into a Pandas DataFrame, and all four feed into a common Random Forest Regression training and evaluation stage, producing console/plot output.

5.2 Low Level Diagram

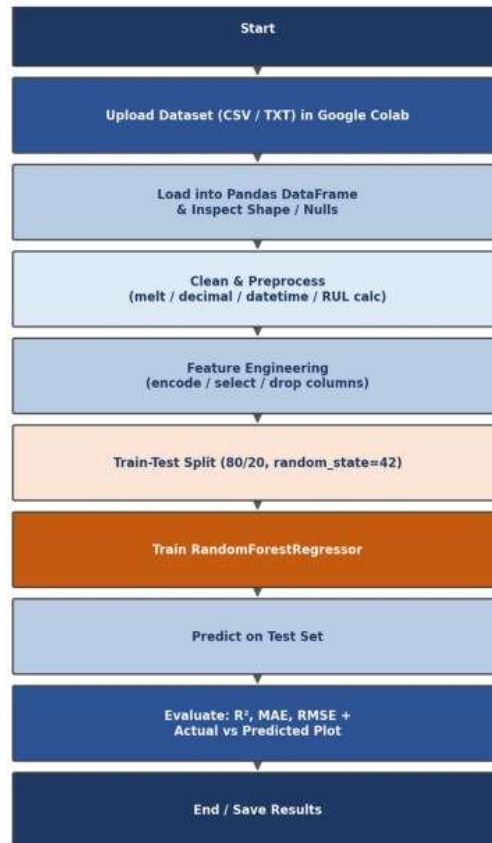


Figure 2: Low-Level Program Flow (Common to All Four Notebooks)

This control flow is common to all four notebooks: dataset upload, cleaning/preprocessing specific to the domain (melt for crop data, decimal handling for mining data, datetime feature engineering for traffic data, RUL calculation for turbofan data), an 80/20 train-test split, Random Forest training, prediction, and evaluation using R^2 , MAE, and RMSE.

5.3 Interfaces



Figure 3: User-System Interface Diagram (Google Colab-Based Interaction)

All four notebooks use Google Colab as the development interface, with dataset upload via the `colab.files` module and console/plot-based output (print statements and Matplotlib scatter plots). No graphical desktop interface or network API is used in the current version.

Module	Core Data / Dataset	Key Logic
Crop Production Prediction	produce.csv (429 × 25, reshaped to 4,272 × 5)	<code>pandas.melt()</code> to convert wide year-columns to long format; Label Encoding; <code>RandomForestRegressor(n_estimators=100)</code>
Mining Quality Prediction	MiningProcess_Flotation_Plant_Database.csv (737,453 × 24)	<code>decimal=","</code> CSV parsing; zip-extract large file; <code>RandomForestRegressor(n_estimators=20, max_depth=15, n_jobs=-1)</code>
Traffic Forecasting	traffic_data.csv (48,120 × 4, engineered to 8 columns)	Datetime feature engineering (Hour/Day/Month/Year/Weekday/IsWeekend); <code>RandomForestRegressor(n_estimators=100, n_jobs=-1)</code>
Turbofan RUL Prediction	train_FD001.txt (20,631 × 26, NASA C-MAPSS)	<code>RUL = max_cycles - time_in_cycles</code> per engine unit; low-signal sensor columns dropped; <code>RandomForestRegressor(n_estimators=100, n_jobs=-1)</code>

6. Performance Test

Since these are independent regression models rather than production-scale systems, the primary performance measures used were the standard regression evaluation metrics — R^2 Score, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE) — computed on a held-out 20% test split for each model.

6.1 Test Plan / Test Cases

Module	Test Case	Expected Result	Status
Crop Production Prediction	Train/test split and fit RandomForestRegressor	Model trains without error; produces R^2 , MAE, RMSE	Pass
Crop Production Prediction	Predict on 855 held-out test rows	R^2 Score ≈ 0.96 , indicating strong fit	Pass
Mining Quality Prediction	Load 737,453-row sensor CSV with decimal=","	Dataset parses correctly with 24 numeric columns, no nulls	Pass
Mining Quality Prediction	Predict % Silica Concentrate on 147,491 test rows	R^2 Score ≈ 0.76 on unseen sensor data	Pass
Traffic Forecasting	Engineer Hour/Day/Month/Weekday/IsWeekend features	New feature columns created correctly from DateTime	Pass
Traffic Forecasting	Predict vehicle volume on 9,624 test rows	R^2 Score ≈ 0.97 , low MAE (≈ 2.4 vehicles)	Pass
Turbofan RUL Prediction	Compute RUL = max_cycles - time_in_cycles per engine	RUL values calculated correctly for all 100 engine units	Pass
Turbofan RUL Prediction	Predict RUL on 4,127 test rows	R^2 Score ≈ 0.72 on multivariate sensor data	Pass

6.2 Test Procedure

Each notebook was executed independently in Google Colab. Datasets were uploaded via the `colab.files.upload()` interface, cleaned and preprocessed, and split into training (80%) and testing (20%) sets using `train_test_split` with a fixed `random_state=42` for reproducibility. After training, predictions were generated on the unseen test set and compared against actual values using `r2_score`, `mean_absolute_error`, and `mean_squared_error` from Scikit-learn. Results were also visually validated using Actual-vs-Predicted scatter plots for each model.

6.3 Performance Outcome

Week	Project	R^2 Score	MAE	RMSE
1	Crop Production Prediction	0.9625	130.83	390.11
2	Mining Quality Prediction	0.7563	0.367	0.555
3	Traffic Forecasting	0.9690	2.399	3.554
4	Turbofan RUL Prediction	0.7173	25.43	35.94

All four models produced the expected output with no unhandled exceptions during execution. Crop Production Prediction and Traffic Forecasting achieved the strongest fit ($R^2 > 0.96$), reflecting clear, learnable relationships between the engineered features and the target. Mining Quality Prediction and Turbofan RUL Prediction achieved moderate R^2 scores (0.72–0.76), which is expected given the higher complexity, noise, and multivariate sensor nature of these two datasets; further feature engineering and hyperparameter tuning are identified as future improvements for these two models.

7. My Learnings

Across the four weeks of this internship, I built a working proficiency in applied Machine Learning while completing four independent, self-contained regression projects.

Week 1 (Crop Production Prediction): Learned to reshape wide-format time-series data using pandas melt, apply Label Encoding to categorical columns, and build a Random Forest Regression model to predict agricultural production, achieving an R^2 Score of 0.96.

Week 2 (Mining Quality Prediction): Learned to work with a very large (737K-row) industrial sensor dataset, including zip-file extraction in Colab and handling comma-decimal-formatted CSV data (decimal=", "), and tuned a Random Forest model (`n_jobs=-1`) for faster training.

Week 3 (Traffic Forecasting): Strengthened my understanding of datetime feature engineering — extracting Hour, Day, Month, Year, Weekday, and IsWeekend features — and applied them to build a highly accurate ($R^2 = 0.97$) traffic volume prediction model.

Week 4 (Turbofan RUL Prediction): Learned to work with the NASA C-MAPSS multivariate sensor dataset, computing Remaining Useful Life (RUL) per engine unit from cycle data, and applied feature selection to drop low-signal sensor columns before training a Random Forest Regression model for predictive maintenance.

Beyond the technical syntax, this internship significantly improved my logical thinking, debugging discipline, and ability to critically evaluate model performance using R^2 , MAE, and RMSE rather than relying on a single metric. Documenting each week's challenges and resolutions also improved my technical communication — an essential skill for any industry role. Overall, this experience has given me stronger confidence in independently designing, building, and evaluating Machine Learning models across diverse real-world domains, and it has directly reinforced the Data Science and problem-solving foundation I am building toward my goal of joining the IT industry.

8. Future Work Scope

- Perform hyperparameter tuning (GridSearchCV / RandomizedSearchCV) on the Mining Quality and Turbofan RUL models to improve their R^2 scores beyond the current 0.72–0.76 range.
- Engineer additional domain-specific features — e.g., rolling averages and lag features for the traffic and turbofan sensor datasets — to capture temporal/degradation patterns more effectively.
- Experiment with additional regression algorithms (XGBoost, Gradient Boosting, LSTM for time-series) and compare their performance against Random Forest Regression.
- Deploy the trained models behind a simple web interface (Flask/Streamlit) so predictions can be generated interactively rather than only within a Colab notebook.
- Extend the Turbofan RUL model to the full NASA C-MAPSS dataset (FD002–FD004, multiple operating conditions and fault modes) for a more generalizable predictive maintenance solution.
- Add model persistence (joblib/pickle) and a proper train/validation/test split with cross-validation for more robust performance estimates.